

Reviewer Pack — Young Tableau Count Exponential Gap in Disjointness Communic...

Entry 2b7d955e959e · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Young Tableau Count Exponential Gap in Disjointness Communication Matrices

Entry ID: 2b7d955e959e **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-11 01:20:07 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Combinatorics (Standard Young Tableau Count) **Field B** (complexity object): Communication Complexity (Disjointness Problem)

Statement:

For a random disjointness instance with n variables, the number of standard Young tableaux of shape λ (where λ is the partition derived from the communication matrix) is exponentially larger than for the corresponding determinant matrix, implying a lower bound on the communication complexity.

Rationale (proposer's reasoning):

The exponential gap in tableau counts reflects the combinatorial structure of the communication matrix, which could be leveraged to derive lower bounds via representation-theoretic methods.

Taxonomy category: COMM_COMPLEXITY (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 72bf5bf044080e6d

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
from math import factorial

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def hook_length_formula(partition):
        n = sum(partition)
        numerator = factorial(n + 1)
        denominator = 1
        for row in partition:
            for cell, size in enumerate(row):
                denominator *= (cell + 1) * (size - cell)
        return numerator // denominator if denominator != 0 else float('inf')

    def determinant_matrix(n):
        return [[random.choice([0, 1]) for _ in range(n)] for _ in range(n)]

    def communication_matrix(determinant):
        n = len(determinant)
        comm_matrix = []
        for i in range(n):
            row_sums = [sum(row[j] for j in range(i + 1)) for row in determinant]
            comm_matrix.append(row_sums)
        return comm_matrix

    def partition_from_matrix(matrix):
        n = len(matrix)
        sums = [sum(row) for row in matrix]
        sorted_sums = sorted(sums, reverse=True)
        partition = []
        current_sum = 0
        i = 0
        while current_sum < n:
            if sorted_sums[i] > 0:
                partition.append(sorted_sums[i])
                current_sum += sorted_sums[i]
                sorted_sums[i] = 0
            else:
```

```

        i += 1
    return partition

def count_standard_young_tableaux(partition):
    return hook_length_formula(partition)

n = random.randint(5, 40)
determinant = determinant_matrix(n)
comm_matrix = communication_matrix(determinant)
det_partition = partition_from_matrix(determinant)
comm_partition = partition_from_matrix(comm_matrix)

det_tableau_count = count_standard_young_tableaux(det_partition)
comm_tableau_count = count_standard_young_tableaux(comm_partition)

ratio = comm_tableau_count / det_tableau_count
conjecture_holds = ratio > 2**n

return {
    "metric_name": "Ratio of Tableau Counts",
    "metric_value": ratio,
    "instances_tested": 1,
    "conjecture_holds": conjecture_holds,
    "counterexample": "" if conjecture_holds else f"Det: {det_partition}, Comm: {comm_partition}"
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [random.getrandbits(32) for _ in range(30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_ratio = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(r["conjecture_holds"] for r in results) / len(results)

    if support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_ratio} std=NA support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results):
        counterexample = next(r["counterexample"] for r in results if not r["conjecture_holds"])
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

```

Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_525950f5.py", line 86, in <module>
    result = run_trial(seed)
              ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_525950f5.py", line 66, in run_trial
    det_tableau_count = count_standard_young_tableaux(det_partition)
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_525950f5.py", line 58, in count_standard_young_tableaux
    return hook_length_formula(partition)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_525950f5.py", line 26, in hook_length_formula
    for cell, size in enumerate(row):
                       ^^^^^^^^^^^^^
TypeError: 'int' object is not iterable

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test crashed due to `TypeError`, preventing data collection. The crash indicates a code bug rather than a conjecture failure. | next: Fix the `hook_length_formula` implementation to handle partitions correctly

11. Audit log (LLM calls)

Total LLM calls: 8

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	112207
2	propose	ollama_remote	qwen3:8b	0	0	96308
3	preregistration	ollama_remote	qwen3:8b	0	0	25009
4	novelty	ollama_remote	qwen3:8b	0	0	20764
5	novelty	ollama_remote	qwen3:8b	0	0	20038
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11157
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8868
8	judge	ollama_remote	qwen3:8b	0	0	18150

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 312501 ms total latency. Provider mix: {'ollama_remote': 8}

(full prompt+response transcripts available in `research/audit/2b7d955e959e.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71

2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/2b7d955e959e.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/2b7d955e959e.tar.gz` (if generated)